# CPS 843/CP8307 Final Project Report

Due Monday, December 8th, at 11:59pm (20 points)

Every student is required to complete a project as an essential component of this course. Project topics are flexible and may be selected at the student's discretion, focusing on areas such as image processing, computer vision, or pattern recognition. Topics may be chosen based on personal interests or aligned with ongoing research initiatives.

## Purpose

The objective of this project is to provide hands-on experience in solving a computer vision problem. Students are expected to develop an in-depth understanding of the chosen problem, gain familiarity with current state-of-the-art solutions, and acquire practical experience by implementing one or more algorithms to address the issue. The project also aims to cultivate fundamental research skills.

## Requirements

- CPS 843: Group work is highly encouraged. Students may work on the project in a team of up to three members. You may also choose to work independently.
- CP 8307: For students enrolled in CP 8307, independent work is mandatory.
- Enroll yourself in a group on D2L by November 3rd, even if you are working independently.

## General Guidelines

- The report is due on Monday, December 8th, at 11:59pm. Late submissions will not be accepted.
- Use IEEE double-column format. The Word and LaTex template can be found at http://www.ieee.org/conferences_events/conferences/publishing/templates.html
- The length of your report should be 4 to 6 pages long, excluding the references. Resize all images properly in line with the text of your report.
- Submit your report in one PDF file through D2L (multiple submission is allowed, but only the last submission will be kept and evaluated).
- Upload all your source code to Google Drive or GitHub and share the link at the end of the Abstract of your report.
- You may use any programming languages, software platforms, and packages for your project. You may directly use the source code shared by the authors or other researchers, but you should give proper references and identify the source in your report.
- If you choose group work, every member should be actively involved and make reasonable contributions to the project. Each group is only required to submit one report; however, you should identify the role and contributions of each member in the project at the end of the Conclusion section.
- Complete the report in your own words. We will use Turnitin® for similarity check.

**General Structure of Your Report**

- Title, author(s), author(s) affiliation
- Abstract: A summary of your report and the main results (add a link to your source code at the end of the Abstract).
- Introduction: Introduce the problem you want to solve and why it is important. Review previous work with proper citations. Briefly describe your method and the main contributions of this work.
- Technical part: Detailed description of your method, algorithms, or models. You can use figures or other visual aids to enhance your explanation.
- Experiments: Present your data and experimental details, results, as well as performance analysis and comparisons with other approaches. Include the parameter settings where applicable. You may present the results using tables, plots, graphs, and other visual aids.
- Conclusion: The main conclusion of your project (identify the role and contributions of each member in the project)
- References: All references cited in your report

**Evaluation Principle**

- Clearly articulate your approach, experimental methodology, and analysis in the report, ensuring that all essential components are thoroughly documented.
- Adhere strictly to academic integrity standards: avoid plagiarism and provide proper citations for all references, including datasets and source code.
- If your project contains novel contributions, I am available to support you in refining and preparing the work for potential publication.
- Successful completion of the project and submission of a well-structured report are mandatory. Evaluation will consider your understanding of the problem, technical explanations, experimental results, and the depth of your analysis.
- Please refer to the project rubric for further guidance and details regarding the evaluation criteria.
- Below is a preliminary breakdown of the project grading criteria:

    - Abstract & introduction 4/20
    - Technical part 6/20
    - Experiments & analysis 6/20
    - Completeness 4/20